SYSC 5108 Exam Study
Assignment 2, Question 4

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Question

For the multivariate logistic regression model from Assignment 2, Question 4, continuous input features are normalized to the range $[-1, 1]$, missing continuous values are imputed with the mean, and missing categorical values are imputed with the mode. Use the trained model to predict whether each query instance will make a repeat purchase.

Course and Textbook Cross-References

- **Chapter 3 slides, Logistic Regression Model:** for $K = 2$ classes, the model uses a linear score $z = \mathbf{w}^T\mathbf{x}$ and converts it to a probability with the logistic sigmoid.
- **Chapter 3 slides, Logistic Sigmoid Function:**

$$\sigma(z) = \frac{1}{1 + \exp(-z)}.$$

- **Chapter 3 slides, Prediction:** the predicted class is selected by comparing class probabilities. With two classes, this is equivalent to predicting repeat purchase when $\sigma(z) \geq 0.5$.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 5:** logistic regression maps a linear function through a sigmoid and interprets $p(y = 1 \mid x; \theta) = \sigma(\theta^T x)$ as the probability of class 1.

Given Information

Data Quality Report

The continuous-feature statistics needed for range normalization and imputation are:

Feature	Min	Mean	Max	1st Qrt	Median	3rd Qrt
Age	18.0	32.7	63.0	22.0	32.0	43.0
Shop frequency	0.2	2.2	5.4	1.0	1.3	4.3
Shop value	5.0	101.9	230.7	11.8	100.14	174.6

The categorical mode for socioeconomic band is

$$\text{mode(socioeconomic band)} = a.$$

Model Weights

Feature	Weight
Intercept w_0	0.6679
Age	-0.5795
Socioeconomic band B	-0.1981
Socioeconomic band C	-0.2318
Shop value	3.4091
Shop frequency	2.0499

Query Instances

ID	Age	Band	Shop frequency	Shop value
1	38	a	1.90	165.39
2	56	b	1.60	109.32
3	18	c	6.00	10.09
4	?	b	1.33	204.62
5	62	?	0.85	110.50

Step-by-Step Solution

1. Impute missing values

Missing continuous values are imputed with the mean, so ID 4 has

$$\text{Age}_4 = 32.7.$$

Missing categorical values are imputed with the mode, so ID 5 has

$$\text{Band}_5 = a.$$

2. Encode the categorical feature

The model uses one-hot coding for bands b and c. Band a is the baseline:

$$a \to (B, C) = (0, 0), \qquad b \to (B, C) = (1, 0), \qquad c \to (B, C) = (0, 1).$$

3. Normalize continuous features

Range normalization to $[-1, 1]$ is

$$x_{\text{norm}} = 2\frac{x - x_{\min}}{x_{\max} - x_{\min}} - 1.$$

For example, for ID 1 age:

$$\text{Age}_{1,\text{norm}} = 2\frac{38 - 18}{63 - 18} - 1 = -0.1111.$$

Assumption for ID 3: the query value Shop frequency $= 6.00$ is above the training-set max 5.4. Since the question says features are normalized to $[-1, 1]$, this solution clips ID 3's shop frequency to the training max 5.4 before normalization. Without clipping, ID 3's normalized frequency would be 1.2308 and its prediction changes; see the exam note at the end.

ID	Age norm	Freq. norm	Value norm	B	C
1	-0.1111	-0.3462	0.4213	0	0
2	0.6889	-0.4615	-0.0756	1	0
3	-1.0000	1.0000	-0.9549	0	1
4	-0.3467	-0.5654	0.7689	1	0
5	0.9556	-0.7500	-0.0651	0	0

4. Compute the logistic-regression score

Use

$$z = w_0 + w_{\text{age}}x_{\text{age}} + w_B x_B + w_C x_C + w_{\text{value}}x_{\text{value}} + w_{\text{freq}}x_{\text{freq}}.$$

For ID 1:

$$\begin{aligned} z_1 &= 0.6679 + (-0.5795)(-0.1111) + (-0.1981)(0) + (-0.2318)(0) \\ &\quad + (3.4091)(0.4213) + (2.0499)(-0.3462) \\ &= 1.4589. \end{aligned}$$

5. Convert the score to a probability

Use the sigmoid:

$$p(\text{repeat purchase} \mid x) = \sigma(z) = \frac{1}{1 + \exp(-z)}.$$

Predict repeat purchase if $p \geq 0.5$.

ID	z	$\sigma(z)$	Predicted class
1	1.4589	0.8114	Repeat purchase
2	-1.1332	0.2436	No repeat purchase
3	-0.1898	0.4527	No repeat purchase
4	2.1330	0.8941	Repeat purchase
5	-1.6453	0.1617	No repeat purchase

Final Answer

Using mean/mode imputation, one-hot encoding, range normalization, and the clipped ID 3 shop-frequency value, the model predicts:

ID	p(repeat purchase)	Decision
1	0.8114	repeat purchase
2	0.2436	no repeat purchase
3	0.4527	no repeat purchase
4	0.8941	repeat purchase
5	0.1617	no repeat purchase

Exam Notes

- The most common mistakes in this question are forgetting to impute missing values before normalization, encoding band a incorrectly, or applying the sigmoid before adding the intercept.
- The score z is linear, but the final probability is nonlinear because of the sigmoid.
- If the instructor expects *unclipped* range normalization for the out-of-range ID 3 frequency value 6.00, then

  $$x_{\text{freq},3} = 2\frac{6.00 - 0.2}{5.4 - 0.2} - 1 = 1.2308,$$

  which gives

  $$z_3 = 0.2832, \qquad \sigma(z_3) = 0.5703.$$

  Under that convention, ID 3 would be predicted as repeat purchase. The clipped convention matches the assignment solution assumption that normalized features should remain inside $[-1, 1]$.